Backpropogation

30 March 2022 13:18

Backprop → Algo → train nn (How)

training

Pre requisite

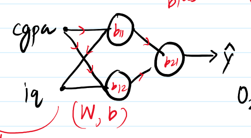
→ Gradient Descent (✓) →
→ Forward Propogation (✓) →

cgpa	iq	lpa
8	80	8
7	70	7

reg

predict

weights, bias → correct value



6+3 → 9 trainable

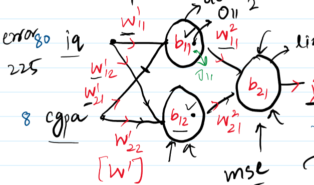
$\sigma(z)$

$O_{21} = W_{11}^2 O_{11} + W_{21}^2 O_{12} + b_{21}$

Backprop

4 students

iq	cgpa	lpa	error
80	8	3	225
60	9	5	
70	5	8	
120	7	11	



L min → $\hat{y}$ → O_{21}

(W, b) Random val

$\boxed{W \to 1, b \to 0}$

Steps →

0) Init w, b

1) You select a point (row) ↳ student

2) Predict (lpa) → forward prop [Dot product]

3) Choose a loss function (mse)

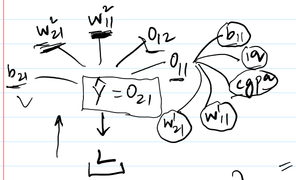
4) Weights and bias update ↳ Gradient Descent

→ $\boxed{18 \text{ lpa}}$

$L = (y - \hat{y})^2$

$(3-18)^2 = 225$ error

Hierarchy ↳ Backprop



$\eta = 1$ (0.1)

$$\boxed{W_{new} = W_{old} - \eta \frac{\partial L}{\partial W_{old}}}$$ ✓

$$\boxed{b_{new} = b_{old} - \eta \frac{\partial L}{\partial b_{old}}}$$ ✓

$$\boxed{W^2_{11\,new} = W^2_{11\,old} - \eta \boxed{\frac{\partial L}{\partial W^2_{11}}}}$$ derivative

$$\boxed{W^2_{21\,new} = W^2_{21\,old} - \eta \frac{\partial L}{\partial W^2_{21}}}$$ ✓

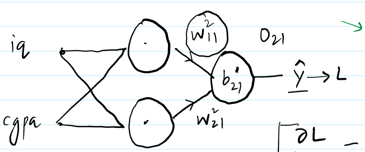
$$\boxed{b_{21\,new} = b_{21\,old} - \eta \frac{\partial L}{\partial b_{21}}}$$ ✓

derivative of loss wrt weight

$\boxed{\frac{\partial L}{\partial W^2_{11}}}$ partial derivative

derivative $\frac{\partial L}{\partial W^2_{11}}$

$\frac{\partial L}{\partial W^2_{11}}, \frac{\partial L}{\partial W^2_{21}}, \frac{\partial L}{\partial b_{21}}$ | $\frac{\partial L}{\partial W^1_{11}}, \frac{\partial L}{\partial W^1_{21}}, \frac{\partial L}{\partial b_{11}}$ | $\frac{\partial L}{\partial W^1_{12}}, \frac{\partial L}{\partial W^1_{22}}, \frac{\partial L}{\partial b_{12}}$

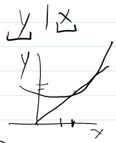


$$\frac{\partial L}{\partial W^2_{11}} = \boxed{\frac{\partial L}{\partial \hat{y}}} \times \boxed{\frac{\partial \hat{y}}{\partial W^2_{11}}}$$ → chain rule of differen

$\hat{y}$, W_{11}^1, L (min)

$\left\{\frac{dy}{dx}\right\}$

y | x



$$\boxed{\frac{\partial L}{\partial \hat{y}}} = \frac{\partial}{\partial \hat{y}}(y - \hat{y})^2 = \boxed{-2(y-\hat{y})}$$

$$\frac{\partial \hat{y}}{\partial W^2_{11}} = \frac{\partial}{\partial W^2_{11}}\left[O_{11} W^2_{11} + O_{12} W^2_{21} + b_{21}\right] = O_{11}$$

$\hat{y} \leftarrow W^2_{21}$, $\frac{\partial L}{\partial W^2_{21}}$, L

$$\boxed{\frac{\partial L}{\partial W^2_{11}} = -2(y-\hat{y})O_{11}} \quad 1$$

$$\frac{\partial L}{\partial W^2_{21}} = \boxed{\frac{\partial L}{\partial \hat{y}}} \times \frac{\partial \hat{y}}{\partial W^2_{21}}$$

$$\frac{\partial \hat{y}}{\partial W^2_{21}} = \frac{\partial}{\partial W^2_{21}}\left[O_{11}W^2_{11} + O_{12}W^2_{21} + b_{21}\right] = O_{12}$$

$$\boxed{\frac{\partial L}{\partial W^2_{21}} = -2(y-\hat{y})O_{12}} \quad 2$$

$\hat{y} \to b$, L

$$\frac{\partial \hat{y}}{\partial b_{21}} = 1$$

-> At first we will perfrom forward propagation (after initializing the weight and bias randomly) on the first row.
-> Then compute the loss by using a loss function.
-> After we get the loss, we will go back to update those weights and biases so that our loss become minimum
-> We are going back to update those weights and biases that's why it is called backpropagation.
-> And we will update those parameter by using the gradient descent approach.

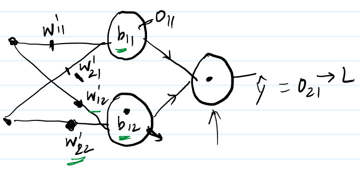
-> next we will calculate derivative of all 9 learning parameter for our example to perform gradient descent.
-> here we use chain rule to perform the derivatives

$$\boxed{\frac{\partial L}{\partial w^2_{21}} = -2(y-\hat{y})O_{12}} \quad 2$$

$\hat{y} \to b$, $\uparrow L$

$$\frac{\partial \hat{y}}{\partial b_{21}} = 1$$

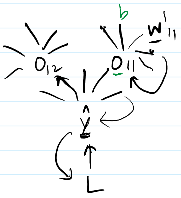
$$\boxed{\frac{\partial L}{\partial b_{21}} = -2(y-\hat{y})} \quad 3 \qquad \frac{\partial L}{\partial b_{21}} = \boxed{\frac{\partial L}{\partial \hat{y}}} \times \frac{\partial \hat{y}}{\partial b_{21}}$$



$$\frac{\partial L}{\partial w'_{11}} = \frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial O_{11}}\frac{\partial O_{11}}{\partial w'_{11}} \to \boxed{-2(y-\hat{y})\, w^2_{11}\, x_{i1}} \quad 4$$

$$\frac{\partial L}{\partial w'_{21}} = \frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial O_{11}}\frac{\partial O_{11}}{\partial w'_{21}} \to \boxed{-2(y-\hat{y})\, w^2_{11}\, x_{i2}} \quad 5$$

$$\frac{\partial L}{\partial b_{11}} = \frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial O_{11}}\frac{\partial O_{11}}{\partial b_{11}} \to \boxed{-2(y-\hat{y})\, w^2_{11}} \quad 6$$



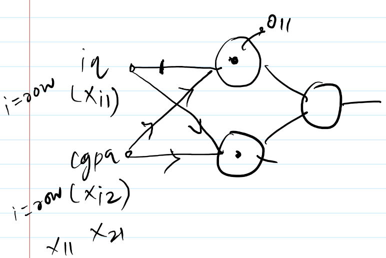
$$\frac{\partial L}{\partial w'_{12}} = \frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial O_{12}}\frac{\partial O_{12}}{\partial w'_{12}} \to \boxed{-2(y-\hat{y})\, w^2_{21}\, x_{i1}} \quad 7$$

w^2_{11}

$$\frac{\partial L}{\partial w'_{22}} = \frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial O_{12}}\frac{\partial O_{12}}{\partial w'_{22}} \to \boxed{-2(y-\hat{y})\, w^2_{21}\, x_{i2}} \quad 8$$

$$\frac{\partial \hat{y}}{\partial O_{12}} = w^2_{21}$$

$$\frac{\partial L}{\partial b_{12}} = \frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial O_{12}}\frac{\partial O_{12}}{\partial b_{12}} \to \boxed{-2(y-\hat{y})\, w^2_{21}} \quad 9$$



$$\frac{\partial O_{11}}{\partial w'_{11}} = \frac{\partial}{\partial w'_{11}}\left[iq\, w'_{11} + cgpa\, w'_{21} + b_{11}\right] = (iq)$$

$\uparrow x_{i1}$

$$\frac{\partial O_{11}}{\partial w'_{21}} = x_{i2}$$

$$\frac{\partial O_{11}}{\partial b_{11}} = 1$$

$$\frac{\partial O_{12}}{\partial w'_{12}} = \frac{\partial}{\partial w'_{12}}\left[iq\, w'_{12} + cgpa\, w'_{22} + b_{12}\right] = iq\,(x_{i1})$$

$$\frac{\partial O_{12}}{\partial w'_{22}} = x_{i2}$$

$$\frac{\partial O_{12}}{\partial b_{12}} = 1$$

multiple chance

epochs

Steps (once again)

0) weights / bias → init → random; w=1, b=0

loop - 100/100

converge

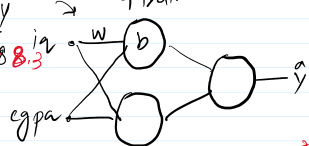
1) for i in range(4):

cgpa	iq	lpa	$\hat{y}$
8	80	8	8.3
6	60	6	
7	70	7	
9	90		

1 stu

9 trainable

$L \downarrow$



$(y-\hat{y})^2$ → Backprop

1a) 1 student → forward prop → lpa) →

12 min

1b) Loss calculate (mse) →

1c) Adjust all weights and bias

The How

4D

$$\boxed{W_{new} = W_{old} - \eta \frac{\partial L}{\partial W_{old}}} \to 9 \text{ times}$$

Backpropagation Algorithm

```
epochs=5
for i in range(epochs):
    for j in range(X.shape[0]):
```

$$\frac{\partial L}{\partial w^2_{11}} = -2(y-\hat{y})O_{11}$$

$$\frac{\partial L}{\partial w^2_{21}} = -2(y-\hat{y})O_{12}$$

Ready